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Key System Applications for the Digital Age

Business Information Systems — Management of Information Systems (IS) Assessment · Week 03 · Student Edition

Enterprise Applications · E-Commerce · Managing Knowledge · Enhancing Decision Making

From the Industry to the Classroom — Building the IT's Future, Today and Together!

How to use this deck (students)

- **Slides are the map; the lecture document is the territory:**
  - Read Week03_Lecture_Key_System_Applications_STUDENT.html (or the .md lecture) for the full 2-hour lecture — it works offline.
- Keep all Week 03 files in ONE folder so diagrams and labs work.
- Run the four lab files as you reach them — every line of code is commented.
- Finish with the 10-question self-check quiz: 8/10 or better means you are ready for the Week 03 lab.

Session roadmap — 120 minutes

- 00:00–00:08 Welcome and objectives
- 00:08–00:38 Lesson 1 — Achieving Operational Excellence and Customer Intimacy: Enterprise Applications
- 00:38–01:06 Lesson 2 — E-Commerce: Digital Markets, Digital Goods
- 01:06–01:30 Lesson 3 — Managing Knowledge
- 01:30–01:52 Lesson 4 — Enhancing Decision Making
- 01:52–02:00 Wrap-up, quiz instructions, lab preview

Learning objectives — six concrete abilities

- Explain ERP (Enterprise Resource Planning) and why one shared database matters.
- Describe SCM (Supply Chain Management) and CRM (Customer Relationship Management) and the goals they serve.
- Explain what makes digital markets and digital goods economically different.
- Distinguish tacit from explicit knowledge; walk the 4-step knowledge value chain.
- Match structured / semi-structured / unstructured decisions to MIS / DSS / ESS.
- Run and modify four small programs (2 × Python, 2 × HTML) that prove each lesson.

LESSON 1

Achieving Operational Excellence and Customer Intimacy: Enterprise Applications

Two goals every business chases

- Operational excellence — everyday work done faster, cheaper, with fewer errors (measurable efficiency).
- Customer intimacy — knowing each customer individually and serving them personally.
- **Both are information problems:**
  - – Efficiency needs every department to know the true current state.
  - – Intimacy needs the company to remember every interaction.

Enterprise =
the WHOLE organization,
not one department

The problem first: information silos

- Each department bought its own software; each holds a private copy of the data.
- **Guaranteed consequence: the copies drift apart.**
    - Warehouse says 40 units; sales already sold 2; accounting used last year's price.
    - Three departments, three different answers to 'how is the business doing?'
- Staff re-type data between systems — every re-typing is a fresh chance for an error.

The cure: ERP (Enterprise Resource Planning)

- **One integrated system built around ONE shared central database.**
- All departments read from it and write to it — no private copies, no re-typing.
- Every number comes from the same place, so every report agrees with itself.
- This agreement is called a single source of truth.

ONE
shared database

What happens on ONE sale (automatically)

- The counter records the sale once. Then, with no re-typing:
  - – Inventory count drops — warehouse and kitchen see true stock instantly.
  - – Revenue is recorded at the CURRENT official price — accounting is correct.
  - – The customer's record updates — marketing and service know what happened.
- **One real-world event → one entry → every department updated.**

ERP modules and the vendors you will meet

- Sold as modules: finance, inventory, human resources, manufacturing, sales — activate what you need.
- Major vendors: SAP (sap.com), Oracle (oracle.com), Microsoft Dynamics 365 (microsoft.com).
- Open-source option: Odoo (odoo.com) — popular with small businesses.

SCM (Supply Chain Management) — excellence beyond your walls

- The supply chain: suppliers → manufacturers → warehouses → transporters → stores → customers.
- **SCM systems share information ACROSS companies so supply matches real demand.**
- ERP organizes information inside one company; SCM extends it along the chain.

Two facts about supply chains

• Bullwhip effect — when chain members do not share real sales data, each adds a 'just in case' margin; small demand changes become large order swings upstream. Remedy: share the true numbers.

• Push vs pull — push: produce on forecasts (guesses) and push toward stores; pull: actual purchases trigger production. Modern SCM moves companies toward pull, because real numbers beat guesses.

CRM (Customer Relationship Management) — the intimacy machine

- One system gathers EVERY interaction with EVERY customer: purchases, emails, support calls, complaints, loyalty points.
  - – Sales sees what each customer owns and might want next.
  - – Marketing sends offers matching a person's actual history.
  - – Service sees your whole story — you never repeat it a third time.

Two CRM numbers managers watch

- Churn rate — the percentage of customers who stop buying in a period. Lower is better.
- CLTV (Customer Lifetime Value) — estimated total profit from one customer over the whole relationship.
- CLTV tells you how much a loyal customer is worth — and therefore how much keeping them can cost.

Churn ↓
CLTV ↑
the whole CRM game

The honest slide: what enterprise applications cost

- Expensive: licences, consultants, training, months of work.
- They force the company to change work habits to match the software — people resist.
- Switching costs: once everything runs on one vendor, leaving is very hard and very costly.
- **Installing the software is the easy half; changing how people work is the hard half.**

Lesson 1 — video and real-life example

- Video (~4 min): 'What is ERP? — A Simple Explanation' — youtube.com/watch?v=10JeksrGVjI (linked only; checked Aug 18, 2026).
- **Example — Poutine Palace, 5 locations:**
  - Silos: location 3 sells out of curds while location 5 discards expired ones; Amira's loyalty points exist at one counter only; month-end closes two weeks late.
  - With ERP + CRM: one shared database — stock, revenue and Amira's points update everywhere on every sale; transfers happen BEFORE the Friday stock-out.

Lesson 1 lab — Week03_L1_ERP_Order_Flow.py

- Simulates ONE sale twice: through one shared database (ERP way) and through disconnected department copies (silo way).
- The silo run 'forgets' the kitchen update and uses an old price — the printed reports disagree.
- Runs offline: python3 Week03_L1_ERP_Order_Flow.py — every line is commented.
- Try it: set the sale quantity to 100 and watch the ERP politely refuse an impossible sale.

LESSON 2

E-Commerce: Digital Markets, Digital Goods

What e-commerce is — beyond 'a store, but online'

- E-commerce (electronic commerce): buying and selling goods or services over the internet, including the marketing and payment around the transaction.
- The lazy definition — 'a store, but online' — misses the point:
  - **a digital market has properties NO physical store can have, and they change the economics of selling.**

What only a digital market can do (1/2)

- Ubiquity — available everywhere, all the time; the store is in your pocket at 3 a.m.
- Global reach — a one-person Montréal shop sells to Osaka with the same click.
- Universal standards — every site speaks the same internet standards, so reaching the market costs a fraction of renting a storefront.

What only a digital market can do (2/2)

- Information richness & density — video, reviews, full specs, instant price comparison; the seller-buyer information gap shrinks (reduced information asymmetry).
- Personalization — the store greets YOU and recommends from YOUR history, for millions at once.
- Interactivity & social technology — customers talk back; reviews become part of the product page.

What changes in a digital market

• Search costs collapse — comparing sellers takes seconds, so sellers compete harder.

• Transaction costs drop — the cost of making the deal happen (paperwork, payment, travel) shrinks.

• Menu costs drop — changing a price = editing a database field → dynamic pricing (prices moving many times per day, e.g. flights).

• Disintermediation — makers sell direct; wholesaler/retailer layers can disappear (but platforms and delivery apps arrive as NEW middlemen taking their cut).

Digital goods and marginal cost

- Digital good — a product delivered as pure data: e-book, song, stream, software licence, PDF (Portable Document Format).
- **Marginal cost — the cost of producing and delivering ONE MORE unit.**
  - – Printed book: paper + printing + shipping, every copy, forever.
  - – Same book as PDF: a few cents of server and bandwidth — for copy 10 or copy 10 million.
- **First copy can be very expensive; every next copy is nearly free, worldwide, instantly.**

one more copy:

$9.50 vs $0.05

printed vs PDF

Who sells to whom — the e-commerce types

- B2C (Business-to-Consumer) — company → person: headphones on Amazon.
- B2B (Business-to-Business) — company → company: 200 kg of flour from a wholesaler's portal.
- C2C (Consumer-to-Consumer) — person → person via a platform: selling your old textbook.
- m-commerce (mobile commerce) — any of the above, on a phone.
- Social commerce — buying directly inside a social network's feed.

How the money is made — revenue models

- Sales — sell the product, collect the price (a $12 PDF study guide).
- Subscription — repeating fee for ongoing access (Netflix, Spotify).
- Advertising — show ads, charge advertisers (YouTube, free news).
- Freemium — free basic tier, paid premium (Duolingo, Dropbox).
- Transaction fee — take a cut of each deal (eBay, Uber, payment processors).
- Affiliate — commission for delivering a buyer ('Buy now' links on review sites).

Two Canadian notes

- Shopify (shopify.com) — a Canadian platform on which one person can open a B2C store in an afternoon.
- PIPEDA (Personal Information Protection and Electronic Documents Act) — Canada's federal privacy law; online stores collecting personal data must respect it.
- Official explanation: Office of the Privacy Commissioner of Canada — priv.gc.ca.

Lesson 2 — video and real-life example

- Video (~7 min): 'What is E-Commerce? Definition, Types, and Business Models' — youtube.com/watch?v=qF1FMTT17tQ (linked only; checked Aug 18, 2026). Backup: youtube.com/watch?v=8zeIv44bF68.
- **Example — Naomi's $12 study guide, two forms:**
  - Printed: $9.50 per copy to deliver → $2.50 profit; 50 copies = $125; France = shipping kills it.
  - PDF: $0.05 per copy → $11.95 profit; 50 copies = $597.50; France pays the same $12, delivered in seconds; 500 buyers = no extra work.

Lesson 2 lab — Week03_L2_Mini_eStore.html

- Interactive simulator of Naomi's store: one 'copies sold' slider, two live product cards (printed vs PDF) with revenue, cost, profit and bars.
- Self-contained HTML — double-click, works offline, zero external libraries, zero CDN (Content Delivery Network) calls.
- Try it: drag from 1 to 1000 — when do the two products stop being comparable businesses?
- Student exercise: set the print cost to $2.00 — when does physical become acceptable again?

LESSON 3

Managing Knowledge

Three words that are NOT synonyms

- Data — raw recorded facts, no context: 2026-08-14, register 3, item 118, $9.50.
- Information — data organized to mean something: 'curd-poutine sales rose 18% at location 3 last week.'
- Knowledge — information + experience + judgment, guiding ACTION: 'that happens on home-game nights — pre-order curds.'
- (+ Wisdom = knowing which knowledge to apply when → the DIKW (Data–Information–Knowledge–Wisdom) hierarchy.)

The distinction the lesson stands on: tacit vs explicit

• Explicit knowledge — written down in transferable form: manuals, procedures, FAQ (Frequently Asked Questions) pages, checklists, commented code. Searchable; survives its author's departure.

• Tacit knowledge — lives in a person's head and hands: the technician who hears a failing fan; the agent who senses which customer needs patience.

• Critical property: tacit knowledge leaves the building whenever its owner does — resignation, retirement, vacation during your worst outage.

The knowledge management value chain — four steps

- 1. ACQUIRE — write down what the expert explains; convert tacit → explicit. (The step everyone skips when 'too busy'.)
- 2. STORE — one organized, backed-up, findable place — not seventeen personal folders.
- 3. SHARE — searchable by the whole team, linked from the ticket screen.
- 4. APPLY — use it on the next real case, and measure that use.
- **Applying reveals gaps → which starts the next round of acquiring. It is a circle, not a line.**

System families that support the cycle

• ECM (Enterprise Content Management) — organization-wide documents and records.

• LMS (Learning Management System) — you are inside one right now (your course platform).

• Wikis and internal knowledge bases — the searchable article collections of daily work.

• KWS (Knowledge Work Systems) — powerful specialist workstations: CAD (Computer-Aided Design) for engineers, VR (Virtual Reality) for training.

Intelligent techniques — and this course's ONE rule about AI

• Expert systems — a specialist's rules stored as if–then statements.

• Machine learning — finds patterns in data without being given the rules.

• LLMs (Large Language Models) — the technology behind AI (Artificial Intelligence) assistants: summarize, draft, natural-language search over everything written down.

• Useful AND risky: they answer fluently even when wrong — fluency is not accuracy.

• The rule: The model drafts. The analyst decides. The analyst's name goes on the record.

Lesson 3 — video and real-life example

• Video (~6 min): 'What is Knowledge Management?' — IBM Technology — youtube.com/watch?v=3_eI5r55XhU (linked only; checked Aug 18, 2026).

• **Example — Marc retires in June:**

 – 22 years of fixes live in his memory; the queue doubles when he is away.

 – Priya runs the chain deliberately: every Marc fix captured as an article → one base → search inside the ticket screen → usage counted.

 – June: printer plays dead, projector goes blank — a first-week student fixes both from articles signed 'Marc'. Marc left; what Marc knew stayed.

Lesson 3 lab — Week03_L3_Knowledge_Base.py

- A miniature help-desk knowledge base: search articles by keywords, capture a new article, count usage per article.
- Each function IS one value-chain step: add_article() acquires, the list stores, search() shares, solve_ticket() applies.
- The demo replays Marc's story: hit → miss-then-capture → hit-after-capture → usage report.
- Runs offline: python3 Week03_L3_Knowledge_Base.py — every line is commented.

LESSON 4

Enhancing Decision Making

Not all decisions are the same animal

- Structured — repetitive, a clear complete rule decides ('unopened return within 30 days with receipt?'). Can be fully automated.
- Unstructured — novel, important, no formula ('expand to Toronto?'). Judgment, values, risk tolerance. Senior management territory.
- **Semi-structured — part calculable, part judgment ('stay open 3 more evening hours?'). Most middle-management decisions live here — and this is where computer support pays off most.**

Herbert Simon's four stages of a decision

- 1. Intelligence — discover and understand the problem (here 'intelligence' = gathering facts). The café's report shows empty evening hours.
- 2. Design — invent and model the options: close earlier / fewer staff / stay open fully.
- 3. Choice — pick one, using the models and numbers.
- 4. Implementation — carry it out AND check whether it worked; feed results back to stage 1.

Match the system to the decision (1/2)

- TPS (Transaction Processing Systems) — record the raw daily events: every sale, every payment. The source of ALL data below; your ERP is built on them.
- MIS (Management Information Systems) — routine scheduled reports: Monday sales summary, low-stock list. Serve STRUCTURED decisions; answer 'what happened?'
- BI (Business Intelligence) — the umbrella term for analyzing business data: dashboards, drill-downs, trends. Tools you will meet: Microsoft Power BI, Tableau.

Match the system to the decision (2/2)

- DSS (Decision-Support Systems) — for SEMI-STRUCTURED decisions. Signature move: what-if (sensitivity) analysis — change an assumption, the model instantly recomputes the outcome. Play with assumptions BEFORE risking money.
- ESS (Executive Support Systems) — for senior management and UNSTRUCTURED decisions: one screen of critical KPIs, internal AND external, drill-down available.
- Balanced scorecard — organize that screen around money + customers + internal processes + learning/growth, so executives never steer by a single number.

Two warnings that keep decision systems honest

• Data quality — a beautiful dashboard fed wrong data produces confident wrong decisions. GIGO (Garbage In, Garbage Out): output quality can never exceed input quality.

  – Lesson 1's 'numbers that agree' is the PRECONDITION for everything in Lesson 4.

• Automated decisions inherit our mistakes — rules and historical data can carry bias, and the system repeats it tirelessly at scale. Monitor and audit; never set-and-forget.

  – Whoever deploys the system owns its mistakes — the analyst's name goes on the record.

Lesson 4 — video and real-life example

- Video (~6 min): 'What is Business Intelligence (BI) and Why is it Important?' — youtube.com/watch?v=uUTMiUuFJdA (linked only; checked Aug 18, 2026).
- **Example — Diego's café, a semi-structured decision through Simon's stages:**
  - Intelligence: ~18 customers/hour, $6.50 spend, $2.25 ingredients, 2 staff at $16.50/h.
  - Design: contribution $4.25/customer; staff $33.00/hour → break-even ≈ 7.8 customers/hour.
  - Choice: what-if runs (half demand? wages $18? one staff?) → 3-hour trial with 2 staff.
  - Implementation: a month of TPS data → next MIS report → back to Intelligence with facts.

Lesson 4 lab — Week03_L4_WhatIf_Dashboard.html

• Diego's model as a mini-DSS: six sliders (customers, spend, ingredient cost, hours, staff, wage) → live revenue/cost bars, computed break-even, GO / NO-GO verdict WITH its reasons.

• Self-contained HTML — double-click, offline, zero CDN.

• Try it: start at GO, then slowly drag customers/hour down — the verdict flips just under 8, exactly the break-even you computed by hand.

• Exercise: find two different settings that both earn ~$100 — which would YOU choose, and what does the model NOT contain?

Wrap-up — four lessons, four sentences

- Enterprise applications (ERP, SCM, CRM) reach both goals through ONE design decision: one shared database instead of drifting copies.
- E-commerce changes selling because digital markets are everywhere at once and one more digital copy costs almost nothing.
- Knowledge management exists because the most valuable knowledge is tacit — it leaves with the person — so acquire, store, share, apply it while you can.
- Decision support matches the system to the decision type — and none of it works without honest data (GIGO).
- **All together: today's systems are how organizations remember, sell, know, and decide.**

The Week 03 lab builds ON today's files

- L1 file: add a second product and a restock() function; refuse restocking a product that does not exist.
- L2 file: add a third card — a SUBSCRIPTION version — showing monthly recurring revenue.
- L3 file: add top_articles(n); make 'wifi keeps dropping' find the right fix.
- L4 file: add a fixed nightly cost (lighting, heating) — watch the break-even move.
- **Extend, do not start from zero — exactly how you will meet code in industry.**

Before the lab: self-check and the package

- Do the 10-question quiz in the lecture document (answer key at the bottom, after committing on paper). 8/10 or better → ready.
- **The Week 03 package — keep in ONE folder:**
  - Lecture .md · 4 code labs (.py/.html) · 3 diagrams (.svg) · workbook (.xlsx) · 4 reference tables (.csv) · this deck (.pptx).
- Everything except the videos works fully OFFLINE.

All external links (checked Aug 18, 2026)

- Videos: youtube.com/watch?v=10JeksrGVjI (ERP) · qF1FMTT17tQ (e-commerce) · 8zeIv44bF68 (backup) · 3_eI5r55XhU (knowledge management, IBM) · uUTMiUuFJdA (business intelligence).
- Vendors & bodies: sap.com · oracle.com · microsoft.com (+ Power BI) · odoo.com · salesforce.com · hubspot.com · zoho.com · amazon.com · shopify.com · ibm.com · tableau.com.
- References: priv.gc.ca (PIPEDA) · nobelprize.org (Herbert Simon).
- **Scan any link first: virustotal.com/gui/home/url.**

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